

ARTHUR BUIS

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École Polytechnique graduate engineer in Applied Mathematics and Computer Science and incoming DTU Master's student, I am seeking a part-time Student Assistant position in AI/Data Science starting September 2026

Education

Technical University of Denmark

Sept. 2026 - Aug. 2028

M.Sc. in Bioinformatics - Lyngby

- Specialized track in **AI for Life Sciences, Computational Biology, and Genomic Data Analysis**

École Polytechnique

Sept. 2023 - Aug. 2026

Engineering Degree, Master's level - Paris

- Top-ranked French engineering university, with admission through nationally competitive examinations
- Track focused on **applied mathematics, computer science and biology**, with a strong interest in **artificial intelligence**
- Relevant coursework includes: **machine and deep learning, computer vision**, Markov chains, statistics, optimization, signal processing, parallel computing, molecular and cell biology
- **GPA: 3.82/4 — CGPA: 3.91/4**

Lycée Aux Lazaristes

Sept. 2021 - July 2023

'Classe préparatoire' - Lyon

- Two-year intensive studies in **Pure Mathematics, Physics and Computer Science** to prepare the highly competitive entrance exams for top French universities
- **GPA: 4.0/4**

Work experience

SharedStudios

Mar. 2026 - Aug. 2026

Computer Vision Engineer Intern - New York

- Developed a real-time, multi-camera panorama stitching pipeline for **immersive video-conferencing portals**
- Integrated **segmentation functionalities** using **OpenCV** and **YOLO** models to handle parallax due to people or moving objects
- Conducted **literature review** of 10+ computer vision papers to design the production pipeline

Arkhn

June 2025 - Sept. 2025

NLP Engineer Intern - Paris

- Deployed NLP pipelines using Large Language Models tailored for **medical text generation** and synthesis
- Integrated advanced models into a **hospital production environment** adhering to healthcare data regulations, using **vLLM**
- Refined conversational agent architectures to optimize **medical data extraction from unstructured clinical records**

French Army - Adapted Military Service Regiment

Dec. 2023 - Mar. 2024

Officer Cadet / Platoon Leader - New Caledonia

- **Led a platoon of 30+ recruits** through their initial military training
- Participated in recruitment missions within tribal communities in New Caledonia

Project work

Computer Vision for Human Motion Analysis

Sept. 2025 - Dec. 2025

- Designed an **automated posture assessment system for fitness exercises** using video data
- Integrated human body keypoints detection using **MediaPipe/OpenCV** and implemented interpolation methods to handle missing keypoints
- **Trained computer vision models** for exercise classification and time warping algorithms for posture assessment

Supervised Classification of Social Media Texts

Sept. 2025 - Dec. 2025

- Built a full ML pipeline to classify social media users into social roles using deep learning models
- Conducted **extensive experimentation, data preprocessing, model selection, and tuning**, to analyse how behavioral signals and learnt representations capture underlying user roles

Construction and Analysis of Similarity Graphs

Mar. 2025 - May 2025

- Constructed similarity graphs between artists in the French hip-hop scene using tabular and textual data collected via web scraping and APIs
- Analyzed network structure using graph theory methods: **community detection, identification of central nodes**, and study of global graph properties

Large Language Models for Code Generation

Sept. 2024 - May 2025

- Studied and **trained large language models** for code understanding and generation in a poorly documented programming language (Envision)
- Designed an automated evaluation protocol based on an **LLM-as-a-Judge** framework
- Implemented a **Retrieval-Augmented Generation** pipeline using **LangChain** framework

Multi-objective Optimization of Urban Trail Routes

Sept. 2022 - June 2023

- Designed an urban trail routing algorithm based on **graph theory and multi-objective optimization**
- Implemented the full algorithm in OCaml and evaluated it on a large real-world dataset (city of Lyon)

Skills

Languages: French (Native) · English (Fluent, C1 – TOEFL iBT: 106/120) · Spanish (Intermediate, B2)

Programming Languages: Python (Advanced) · OCaml · C · Java · JavaScript · SQL · HTML/CSS

Tools & environments: PyTorch · Tensorflow · Scikit-learn · LangChain · OpenCV · Git · GitHub · Docker · Linux (shell) · Hugging Face · vLLM · Slurm · MPI · OpenMP

Outside interests

Associative Involvement: Vice-president of *Atelier Des Ondes*, an association responsible for organizing concerts and managing sound for events at *École Polytechnique*

Music: Rock/pop drummer, regular experience performing on stage and at live concerts

Mathematics Competitions: Two-time winner of the national Archimède competition (1000+ participants)

Sports: Competitive running, participation in 20+ trail and urban races

Tutoring: Private lessons in mathematics and computer science for high school and preparatory class students